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Date:
March 12, 2025

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Executive Summary: Trustworthy AI Systems

VAIL (Verifiable Artificial Intelligence Layer) is a technology company developing open-source infrastructure that enables verifiable and trustworthy AI systems. Our response to this RFI outlines priority policy actions that would advance America's AI leadership by creating a foundation of trust and security for AI applications via a framework that promotes cryptographic verification methods.

Our proposals ensure that AI systems can be trusted as they proliferate across critical applications in national security, healthcare, and finance. Our recommendations focus on developing technical infrastructure, AI development standards, and incentive structures that would enable organizations to verify the authenticity, integrity, and behavior of AI systems with mathematical certainty.

Present Day Threats

The AI industry is experiencing exponential growth, with thousands of new models being developed each month. This rapid proliferation creates several critical security and trust challenges:

1. **Model Authenticity**: How can users verify that they are using the specific AI model they intend to use, rather than a modified or compromised version? Subtle changes to a model's design made by a bad actor can have significant

impact when the model's precise output informs highly consequential decisions such as where to deploy military resources or when to grant access to sensitive data. Due to the highly complex nature of modern AI systems, it is not feasible to detect such model changes through manual inspection.

2. **Claims & Disclosures**: Even if a user is confident they are using the intended version of a model, how can they verify that the model does indeed have the properties it is advertised to possess? Model cards that describe model properties and test results have become a common "nutrition label" for providing assurances of model specs and performance; however, given the already-rapid pace of model development that we are striving to accelerate, FDA-level regulatory scrutiny would not be advisable. Rather, verification technology can be used to validate model card claims in order to ensure accuracy and combat potential errors or misinformation.

3. **Training Integrity**: How can users verify that a model's training process was not compromised to introduce biases or vulnerabilities? Traditional security measures focused on model inputs and outputs may not detect when a training data pipeline has been tampered with since the model's behavior appears normal in most cases. Furthermore, this compromise may remain undetected until a critical moment.

4. **Result Validation**: How can users verify that AI-generated outputs truly came from the expected model and input data? A user may be working with a model that a trusted third party is hosting remotely, but how does the user know that the model result coming back to them was not intercepted and modified in transit?

5. **Secure Model Collaboration**: As AI systems increasingly work together, how can we ensure secure communication and verification between models? With the rise of AI agents, system-to-system interaction will become more ubiquitous, more fully automated, and more normalized. As this happens, agents will gain access to more sensitive information. A single vulnerable agent in a workflow threatens the integrity of the entire system, potentially leaking sensitive information unbeknownst to the user.

These challenges are particularly acute in high-stakes environments where AI systems influence critical decisions affecting human lives, economic stability, or national security.

The Role of Verifiability

Verifiable AI is based on cryptographic techniques that allow one party to prove to a relying party that a data computation or transformation was performed correctly. In the context of AI, these techniques include:

- **Assurance of training data pipeline integrity:** Verifying provenance of training data and how the data was processed before being fed into a machine learning model for training.
- **Model fingerprinting**: Creating a unique model identifier for a given AI model that allows the user to verify that they are using the expected model without needing to access the model's proprietary details.
- **Training verification**: Generating cryptographic proofs that a model was trained on the expected data.
- **Inference verification**: Producing cryptographic evidence that a specific output was genuinely generated by running a specific input through a specific model.

These tools use both hardware and software technology to provide assurances analogous to those that SSL protocols give the internet, or that digital signatures give documents, but in the more complex setting of AI systems.

Priority Policy Actions

We first describe a framework, and follow that with proposed mechanisms for accelerating and bolstering it. This framework would function similarly to how the National Institute of Standards and Technology (NIST) has established cybersecurity frameworks that are widely adopted across sectors.

1. Establish a National AI Verification Framework

We recommend the development of a national framework for AI verification built on common standards, protocols, and infrastructure for verifiable AI. This would include:

AI Verification Standards

These standards would delineate the responsibilities and obligations between model developers and model users. They would also define requirements that systems of proof for various use cases must meet. These criteria should not be prescriptive in terms of technology, but rather should make clear the functional properties that the technology must possess.

Open Reference Implementations

Developing and advocating for openly available software tools that implement verification tooling, which can be adopted by both government and industry. Open-source tooling enables accelerated development from technical experts, as well as customizable capabilities and fully transparent functionality for government security reviews.

Verifiable benchmarking of model performance

To foster trust and enable informed decision-making, AI model producers should establish and adhere to consistent performance standards. These standards would allow users to select the most suitable model for their specific use case. Additionally, in order to build trust, model producers should provide verifiable proof of performance to substantiate claims that their models meet established benchmarks

National Registry for AI systems to store verifiability information

A publicly accessible registry should be established to serve as a definitive source of record for machine learning models used in AI systems. This registry would include model fingerprints, metadata, performance metrics, and verifiable proofs of the computations used to generate this information.

2. Create Verification-Focused Research Programs

We recommend the establishment of dedicated research programs focused on advancing verification technologies for AI. Though verifiable AI technologies have seen significant progress in recent years and deployable solutions are currently available, further research and development are required to support use cases that involve the most advanced AI systems. These programs should address:

Computational Efficiency: Current verification methods require significant computational resources. Research is needed to reduce these requirements and make verification more easily implementable for large-scale AI systems. As AI systems continue to make gains in computing efficiency, so too must accompanying verification technology.

Cryptographic Algorithm Development: While current cryptography systems can generate verifiable proofs of model training and results, their underlying methods require optimization to feasibly run on highly complex AI systems. This optimization will come from both implementing the latest advances from research institutions and continued research into more efficient algorithms for future implementation.

Hardware-based approaches for Trusted Execution Environments: Enhancing hardware solutions specifically designed to expedite verification processes, while preserving the integrity of the data's original processing environment. This presents opportunities for advancements in proof of execution within these environments and increased efficiency in hardware deployment.

Organizational Integration: There is currently a lack of best practice around how to integrate verifiability capabilities into organizational processes and governance structures. We need to close the gap between what is technically feasible and what organizations are able to implement.

These research programs should engage both academic institutions and industry partners to ensure practical applications of the latest verification technology.

3. Develop Incentive Structures for Verifiable AI Adoption

To accelerate the adoption of our proposed framework for verifiable AI systems, we recommend creating the following incentive structures.

Federal Procurement Preferences: Give preference to AI systems with verification capabilities in federal procurement processes.

Tax Incentives: Provide tax benefits for organizations that implement verifiable AI systems in high-risk applications.

Regulatory Streamlining: Create expedited regulatory pathways for AI systems that implement robust verification capabilities.

Grant Program: To further encourage the development of verifiable AI technologies, we recommend establishing a grant program that provides funding for research and development projects in this area. This program would be open to academic institutions, non-profit organizations, and private companies.

These incentives would help offset the initial costs of implementing verification capabilities while creating market demand for continued development of these features.

4. Creation of Office for AI Authenticity

We recommend creating a government office dedicated towards the implementation of the aforementioned policy. The office's responsibilities would include:

- Development of partnerships with leading industry players and research institutions, focusing on government consulting, use case scoping, and solution procurement
- Managing implementation of verifiable AI systems
- Broadening general awareness through marketing campaigns targeted at business leaders as well as the public
- Leading international efforts to ensure seamless interoperability of verifiability solutions among allied nations

This office would have member private and public member organizations working to prioritize its initiatives.

Implementation Approach

We recommend sequencing these policies as follows:

Phase 1: Form internal office with founding members; establish details of the verification framework and research initiatives, and design incentive programs.

Phase 2: Implement incentive programs, launch research initiatives, and build the registry.

Phase 3: Test and deploy initial public infrastructure, and begin requiring verification capabilities for high-risk government AI applications.

Phase 4: Scale verification capabilities to all government AI systems, fully deploy public infrastructure, and establish international agreements on verification standards.

Conclusion

Advancing America's AI leadership requires not just investing in raw AI capabilities, but also in the verification infrastructure that will enable these systems to be deployed with confidence in high-stakes environments. By establishing standards, infrastructure, and incentives for verification, the United States can create a competitive advantage in trustworthy AI while addressing significant risks associated with unverified AI systems. This innovation offers safeguards for both users of these systems and the general American public who are directly and indirectly affected by them.